PROJECT TRICORDER Mk I

Theory of Operations and Design Analysis

How an offline multimodal field instrument was assembled on a Snapdragon 888, why the parts already existed, and why they had not been shipped as one appliance.

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1. Abstract

Project Tricorder Mk I is a sideloaded Android instrument that orchestrates several already-public on-device models and public data feeds behind one operator face. Chat and vision run as llama.cpp OpenAI-compatible servers in Termux. Image synthesis runs as stable-diffusion.cpp (SD Turbo). Speech out is Kokoro ONNX. Speech in is Android SpeechRecognizer. Scene audio is YAMNet in-process; bird species escalate to BirdNET TFLite; flora/fauna use Coral iNaturalist MobileNet on a frozen camera frame. Translation is ML Kit. Space objects use Celestrak TLEs and satellite.js SGP4. Space weather is NOAA SWPC plus HamQSL. Earth weather is NWS. Watches poll NWS CAP and OpenSky. The contribution is not a new foundation model. It is the routing, duty-cycle, Hold semantics, logging, and operator sliders that turn those parts into a field book that still works in airplane mode after first-run fetches.

This paper states how each path actually runs, which controls the operator was given and why, and why cloud vendors and single-purpose apps left this combination on the table. Claims of certification, air-data, or operational HF circuits are explicitly rejected.

2. Design intent

The design target was a Mark I field appliance: point, listen, ask, translate, imagine a referent, log the find with UTC and GPS, and optionally push a short alert to a second unit. The metaphor is Star Trek instrumentation, not a chatbot skin. That forced three constraints that consumer AI apps usually drop: (1) concurrent backends on distinct ports so chat, vision, and Imagine do not evict each other; (2) Hold / continuous / duty-cycle controls so neural TTS is not flooded; (3) an append-only JSONL book rather than a disposable chat transcript.

Reference hardware was a Galaxy Z Fold3 (Snapdragon 888, 12 GB). That class can run a 4B instruction GGUF, SmolVLM2-500M, SD Turbo at 384–512 with 4–8 steps, and Kokoro, if the operator accepts seconds-to-minutes of wall time. It cannot run a frontier multimodal cloud model offline. The instrument is honest about that trade.

3. Process architecture

3.1 Split: APK face vs Termux engines

The APK is a Kotlin WebView shell plus a JavascriptInterface bridge. In-process work (classifiers, translator, location, Nearby, SWPC/NWS fetch, horizon) lives in the APK because TensorFlow Lite and ML Kit already ship as Android libraries and do not need a Unix userspace. Multi-gigabyte GGUF / SD / Kokoro weights do not belong inside an APK: Play and sideload size, update tax, and license provenance all argue against it. Those engines run as long-lived HTTP servers under Termux (F-Droid), launched by ~/bin/tricorder on 8080 (Gemma chat), 8081 (SmolVLM), 8188 (sd-server), and 5000 (Kokoro). The face treats them as localhost OpenAI-shaped endpoints.

This split is the same pattern a laptop user already knows as “Ollama plus a frontend,” adapted to Android’s two-world reality: ART for sensors and UI, Linux userland for compiled C++ inference. Termux compilation of llama.cpp and stable-diffusion.cpp on-device was part of making the unit real, not a demo of cross-compilation only.

3.2 Ports and health lamps

Four lamps (CHAT, VISION, IMAGE, VOICE) probe /health or the root of each port on a timer: green reachable and idle, yellow in-flight, red down. TRANSLATE is a fifth lamp for the ML Kit loop. The operator is taught to wait for green before speaking again. That is an instrument convention, not a chatbot spinner.

3.3 Storage

Field events append to files/logs/field.jsonl with kind, UTC, optional lat/lon, text, and optional JPEG or video path. Models land in files/models/. TLEs in files/tle/. SOL snapshots in files/sol/snapshot.json. ML Kit language packs live in the Play/ML Kit cache. Nothing in that map is a durable archive; the operator is told to export when the survey ends.

4. Theory of operations by path

4.1 Inquiry (Gemma 3 4B-IT GGUF)

Mic → Android SpeechRecognizer → text box → local tuple match → else POST /v1/chat/completions on :8080 → strip periods that break Kokoro on abbreviations → POST :5000 for WAV → MediaPlayer. Tuples intercept HELP and the COMPUTER easter egg before the LLM. Imagine keywords (show me / generate / picture of) divert to :8188. The query is scrolled on a phosphor LED matrix with elapsed think time. This path is the only one that needs a 4B-class model; everything else is smaller or non-generative.

4.2 Sensor and Sentry (SmolVLM2 + flora)

The rear camera is an environment stream. Sensor sends a JPEG to :8081 (OpenAI image_url chat). Sentry compares frames for motion, then captions. Hold is a toggle, not a shutter: red freezes the frame and runs flora/fauna TFLite on that bitmap; green resumes. After a Sentry hit the face forces Hold so the stack is not re-describing the same scene every few seconds while Kokoro is still speaking. Continuous machine vision is an explicit opt-in that writes to the CRT only and does not call TTS. That pair of controls is the difference between a usable instrument and a talking strobe.

4.3 Imagine (SD Turbo) and why sliders exist

sd-server on :8188 serves an OpenAI images body. SD Turbo is a 1–4 step checkpoint; the CLI default of ~20 steps was wasted compute and produced the first slow runs. Operator sliders expose size (256 / 384 / 512) and steps (typically 4–8) with cfg-scale 1. Too few steps at 256 yielded color tiles; 4 steps at 512 recovered structure at a cost of minutes on an 888. The sliders exist because there is no free lunch on this SoC, and the operator—not a hidden preset—owns the quality/time trade. The intended field use is one referent picture beside a translated sentence when there is no shared noun and no clip-art binder, not a desktop art studio.

4.4 Speech out (Kokoro)

Kokoro 82M ONNX, the same family PocketPal ships, runs as a small Python server on :5000. Phonemizer + a Termux espeak library stub were required because stock kokoro-onnx assumes glibc dlinfo. Periods in USS and pressure readouts truncated utterances; a pre-strip pass was added. Audio Sense holds YAMNet while a WAV is playing so the loudspeaker is not classified as Speech and does not talk over a watch briefing. Playback is whole-utterance, not streamed—the same limitation as a stock ONNX session on this device.

4.5 Audio sense, YAMNet, BirdNET

YAMNet is a public AudioSet classifier (521 classes) running in-process via TensorFlow Lite Task Audio. It is not "smarter than the operator's ears." It is a timestamped logger and a remote sentry: duty-cycle sliders set listen window and sleep so a tent unit can wake 5 s every 15 s. When a bird-like YAMNet label is high, BirdNET TFLite (Cornell / BirdNET-Analyzer mobile graph) runs on the same clip and writes species + confidence + UTC + GPS to the field book. That escalation exists so the cheap generalist does not pretend to be a species catalog, and the catalog does not run on every footstep.

4.6 Flora and fauna

Coral's public iNaturalist MobileNet TFLite packs classify a still frame. They need the JPEG, not the SmolVLM sentence. Early builds only classified after Hold; later builds also classify the Sensor caption frame. The vision model's prose ("a giant leaf") is not an input to the classifier. If the operator expected flora to fire on text alone, that was a wiring error and was corrected.

4.7 Translator

ML Kit on-device Translation with downloadable pairs (≥50 languages in the catalog). SOURCE and TARGET are wheels, not sliders. AUTO green swaps direction after each completed line. Silence (onEndOfSpeech) triggers translate then device TTS. The recognizer must be destroyed and re-created when leaving the mode; a tight retry produced beep loops. Packs are fetched on earth-link init for en+es only; "all languages" is gigabytes and is opt-in.

4.8 Horizon, G, speed

DeviceMotion gravity is remapped for a portrait hold (face toward operator). Pitch/roll from atan2. Load factor G = |a|/9.80665. SPD is LocationManager ground speed in m/s and km/h, not indicated airspeed. The tape is a curiosity. It is not FAA/NASA certified and is labeled as such on the face and in the manual.

4.9 Satellites and stars

Celestrak GP/TLE groups (stations, amateur Part 97, Iridium NEXT, Globalstar, Orbcomm, GEO, GPS ops, visual) download at init. satellite.js 4.x SGP4 in the APK computes look-angles at the last fix. Polar plot, elevation ≥ 0. Star page is ~40 bright catalog stars, RA/Dec → alt-az, rotated by compass heading. Not Stellarium, not radar.

4.10 SOL SYSTEM

Three-press cycle. (1) Schematic Mercury–Saturn mean-element orbits plus NOAA SWPC Kp / GOES X-ray / SFI-SSN and HamQSL grouped HF day/night, VHF aurora, UHF line derived from Kp. (2) Scrollable NWS pane: six hours + today and next two days; no map. (3) Off. Feeds: services.swpc.noaa.gov, hamqsl.com/solarxml.php, api.weather.gov. HamQSL names bands 80m-40m not 80m; the first parser missed that and printed dashes.

4.11 Watches and mesh

While Audio is armed and a network path exists, ContextWatch polls NWS alerts, OpenSky states in a bounding box, and civic CAP + USGS M≥3.5. Hits log, speak, and optionally push JSON over Nearby Connections P2P_STAR. Media is not streamed. Waze has no public keyless feed and is not polled. "Citizen" on the face is IPAWS/USGS, not the Citizen Inc. product.

5. Operator controls that made it work

The first builds failed in use, not in demos. Fixes that mattered:

Hold after Sensor/Sentry so vision and Kokoro stop looping. Continuous vision as a separate button that never speaks. Duty-cycle sliders on Audio so a camp unit is a logger, not a 100% radio. Imagine size and step sliders after 20-step Turbo runs burned minutes for no quality. Keyboard-pop toggle because the operator lives on the mic. Horizon hide toggle so Audio text is not covered by the artificial horizon. Watch speech holds YAMNet. Translator destroy/re-create on mode switch. Earth weather moved off a fixed canvas onto a scroll pane. Backend lamps so “all red, nothing works” is distinguishable from a JS parse break.

Those are instrument controls. Chat apps optimize for one box and a send key. This face optimizes for not drowning the operator or the SoC.

6. Why the pieces existed and the appliance did not

6.1 The parts were public

llama.cpp + GGUF, SmolVLM, SD Turbo, Kokoro ONNX, YAMNet, BirdNET, Coral iNaturalist, ML Kit Translate, Celestrak, SWPC, HamQSL, NWS, OpenSky, and Nearby Connections were all documented and independently shippable before this project. PocketPal already put GGUF chat and Kokoro on a phone. Sky Map already did stars. Heavens-Above / Gpredict already did SGP4. Google Translate already did offline packs. Ham dashboards already painted HF tables. Termux users already compiled llama.cpp on-device.

6.2 Integration tax vs product tax

Each of those products is a coherent SKU. Stitching them is an integration tax: four local servers that die under SSH, TFLite ABI and Task API mismatches, phonemizer-on-Android, WebView file:// geolocation, SpeechRecognizer busy after mode switch, Play-incompatible APK size if weights are embedded, and a UX that looks like a prop until Hold exists. That tax does not appear in a quarterly roadmap for a chat app, a translate app, or a sky app. It only appears if the requirement is “one face, one log, airplane mode after fetch.”

6.3 Cloud economics

Microsoft, Google, and OpenAI monetize tokens, seats, and data-center utilization. An appliance that is finished after a sideload produces no token stream. On-device Nano / AFM cores exist to cut cloud cost and latency for OEM features (summarize this screenshot, smart reply), not to replace the cloud SKU. Google gates Gemini Nano on AICore and RAM class; Pixel 9a’s smaller Nano drop lost multimodality. Apple routes hard work to Private Cloud Compute and, for the heaviest 2026 tier, third-party GPUs. That is a rational hybrid. It is the opposite of a field instrument that must keep working when 8.8.8.8 does not.

6.4 Liability and certification theater

A vendor that ships an “artificial horizon” and “HF forecast” on a phone inherits implied-nav and implied-safety questions. The cheap answer is not to ship the combination, or to bury it behind “not for navigation.” This project prints that warning in red and still ships the tape, because the operator asked for a demo of IMU fusion and ham weather on the same glass. A public company legal team will usually kill that slide.

6.5 Store policy and model weight gravity

Gemma 4B, SD Turbo, and Kokoro together are gigabytes. Play distribution, update bandwidth, and review policy push vendors toward cloud or toward a single small Nano. Sideload + Termux is how this unit exists. That path is unacceptable as a consumer SKU for the firms that own the stores.

6.6 What this project is not

It is not a new model. It is not faster than PocketPal at TTS. It is not more complete than Stellarium at stars or Gpredict at passes. It is an orchestrator with a field book. The “why not before” answer is that orchestration across domains is a hobbyist and lab problem, and the labs that could fund it are paid to keep the interesting weights in the cloud.

7. Competitive feature matrix

Capability	Tricorder 0.6.8	PocketPal	Gemini Nano / Pixel	Apple Intelligence	Google Translate	Sky Map / Gpredict
Offline LLM chat	Gemma 4B Termux	Many GGUF	Nano, device-gated	~3B on-device + cloud	No	No
Vision caption	SmolVLM2 :8081	Some multimodal GGUF	Nano multimodal on flagships	On-device + PCC	Camera translate	No
Image gen on device	SD Turbo sliders	No	Cloud / limited	ADM cloud tier	No	No
Neural TTS	Kokoro :5000	Kokoro / others	System	System	System	No
YAMNet + BirdNET	Yes, duty cycle	No	No	No	No	No
Flora/fauna TFLite	Coral iNat	No	No	No	Lens (cloud/hybrid)	No
Offline MT ≥50	ML Kit	No	Partial	Partial	Yes	No
Horizon + G + GPS SPD	Yes, uncertified	No	No	No	No	No
SGP4 sat plot	Selected groups	No	No	No	No	Gpredict / HA yes
Star map	Major stars	No	No	No	No	Sky Map full
SWPC + HF bands	Yes	No	No	No	No	Ham sites yes
NWS hourly + 72 h	Yes, scroll	No	Assistant cloud	Weather app	No	No
Field JSONL + mesh	Yes	No	No	No	No	No
Works after airplane mode	After fetches	Yes	Feature-gated	Hybrid	After packs	Yes (catalog)
Certified nav / flight	No	No	No	No	No	No

8. Quadrant (Gartner-style)

Axes: horizontal = locality (cloud-first → offline-after-fetch). Vertical = orchestration breadth (single function → multi-domain instrument). Quadrants are descriptive, not a licensed Gartner Magic Quadrant.

Quadrant	Who sits there	Reading
Leaders (offline × broad)	Tricorder Mk I as an integration, not a company	Broad face, local engines, ugly distribution. Not a market leader in revenue.
Visionaries (cloud × broad)	Gemini app, ChatGPT app, Copilot+ PC visions	Many modes, default path is the data center. On-device is a cache.
Challengers (offline × narrow)	PocketPal, Termux+llama.cpp, Sky Map, HamQSL dashboards, ML Kit Translate	Excellent at one job. Do not log a bird, a leaf, a pass, and a sentence in one book.

Niche (cloud × narrow)	OpenSky viewers, Heavens-Above web, NWS web, Citizen-style alert apps	Best-in-class feeds, no offline instrument contract.

No public vendor occupies the top-left with a supported SKU: offline-after-fetch and broad orchestration. That empty cell is the observation this project is about. It is empty because of §§6.2–6.5, not because SGP4 or YAMNet were secret.

9. Limits and honest failure modes

Imagine is minutes, not interactive. Kokoro is whole-file playback. Flora packs are coarse taxa, not a botanist. OpenSky rate-limits. NWS is CONUS-centric. SOL orbits are mean elements. Horizon is not an ADI. Mesh is JSON only. Termux and weights are the operator's problem. JS parse errors have taken the whole face down (0.6.0). Those are recorded in the operations manual, not hidden.

10. Conclusion

Mk I works because the models were already small enough, the feeds were already public, and the operator accepted Termux, sliders, and warnings. It is a laboratory appliance and a field book. It is not evidence that a Snapdragon 888 replaced a data center. It is evidence that the missing product was an orchestrator with Hold, a duty cycle, and a log—not another chat window.

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